RADON: Robust Autoencoder for Unsupervised Anomaly Detection

Naji Najari
Orange Labs, LIRIS CNRS
Meylan, France
Email: firstname.lastname@orange.com

Samuel Berlemont, Grégoire Lefebvre
Orange Labs
Meylan, France
Email: firstname.lastname@orange.com

Stefan Duffner, Christophe Garcia
LIRIS CNRS, INSA Lyon
Villeurbanne, France
Email: firstname.lastname@liris.cnrs.fr

2021 14th International Conference on Security of Information and Networks (SIN) | 978-1-7281-9266-6/21/$31.00 ©2021 IEEE | DOI: 10.1109/SIN54109.2021.9699174

***Abstract*—Anomaly detection is a critical element in the design of secure and reliable networks. Networks are vulnerable to diverse anomalies, including malicious attacks, traffic congestion, operational problems, and hardware failures. In addition, novel unknown anomalies are constantly emerging, which challenges existing knowledge-based anomaly detectors.**

In this paper, we propose RADON, a robust algorithm for unsupervised anomaly detection, which does not require anomaly signatures or any labeled training data. RADON consists in a novel robust training strategy applied to an autoencoder. First, a filtering rule is applied on the reconstruction scores of traffic metadata to identify potential anomalies contaminating the training data. Then, RADON simultaneously learns to minimize the reconstruction scores of nominal instances, and to maximize that of the filtered anomalies. Finally, the trained model is leveraged to detect anomalous observations. Extensive experiments on the MNIST, NSL-KDD and MedBIoT benchmark datasets illustrate the effectiveness of our approach for anomaly detection, both in image and network traffic analyses.

***Index Terms*—Unsupervised anomaly detection, robust autoencoders, dynamic outlier filtering, network intrusion detection, network security.**

I. Introduction

Nowadays, the proliferation of communication technologies has contributed to developing more and more complex large-scale networks. With the emergence of the Internet of Things (IoT), connected devices have become ubiquitous in several IT systems, including Smart Home, Healthcare, Industry 4.0. Cisco predicted that the number of connected devices will reach 500 billion devices in 2030 [1]. While IoT has contributed to the expansion of new services, it introduces new security challenges. Indeed, many IoT devices run outdated firmwares, rely on old legacy protocols, and have constrained computational resources. These devices are prone to failure and are vulnerable to a wide spectrum of anomalies, such as malicious attacks, traffic congestion, connectivity problems, and flash crowds.

Moreover, recent years have witnessed a significant increase in the occurrence of new unknown anomalies, e.g., zero-day attacks [2]. These anomalies usually go under the radars of traditional network security solutions, namely signature-based Intrusion Detection Systems (IDSs), a.k.a., misuse detectors [3]. Such systems track down predefined signatures of known vulnerabilities in the network traffic, and they raise alarms when the current network behaviour matches one signature. Even though signature-based IDSs detect known attacks with high precision, they fail to detect unknown vulnerabilities and non-malicious anomalies. The continuous update of the signature-based IDS database can be costly and time-consuming.

Unsupervised Anomaly Detection (AD) algorithms are capable of mitigating these limitations. Anomaly-based methods define an anomaly as "an observation that deviates so significantly from other observations as to arouse suspicion that it was generated by a different mechanism" [4]. In essence, anomalies are non-conforming observations that violate an expected nominal behaviour. Firstly, AD builds a model that represents the normal expected network behaviour. Then, any deviation of the current behaviour from the previously built model is deemed anomalous.

AD performance depends on the ability to model normal data patterns through data-driven approaches. Deep neural networks have received great attention in recent years. Especially, AutoEncoders (AE) [5] have become a staple for learning non-linear representation of complex high-dimensional data. Firstly, AE-based anomaly detectors train an autoencoder to reconstruct the nominal network traffic data. Secondly, the reconstruction scores of AEs are used as anomaly scores: the higher the reconstruction score, the higher the probability that the observation is anomalous. Nevertheless, the efficacy of AD approaches depends on the availability of anomaly-free nominal training data. Guaranteeing that no anomalies are infiltrated in the training traffic can be laborious, if not impossible, which limits their applicability in real-world environments.

To our knowledge, few studies have focused on learning a robust representation of the norm, relevant even when the training instances may contain an unknown ratio of unlabeled outliers. To address this problem, we introduce RADON, a Robust Auto-encoder with Dynamic Outlier filteriNg, that learns a robust representation of the nominal class. This approach divides the training data into two groups: a majority group, representing the nominal data that can be accurately reconstructed, and a minority of outliers that are more difficult to reconstruct. Our approach comprises three contributions:

1) an automatic unsupervised filtering of training outliers

978-1-7281-9266-6/21/$31.00 ©2021 European Union

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

through the unimodal thresholding of the reconstruction error histogram;
2) a new formulation of the optimization problem, which not only learns to reconstruct nominal data but also to maximize the reconstruction error of training outliers;
3) an empirical evaluation on three real-world public datasets, which shows that our approach outperforms existing unsupervised anomaly detectors.

This paper is organized as follows: Section II introduces related work in AD, and focuses particularly on Robust Autoencoders. Section III presents our contributions. Section IV depicts the datasets used in our experiments, the training protocols, and the experimental results. Finally, the conclusion and future perspectives are drawn in the last section.

II. Related Work

Anomaly detection (AD) is an active research area that accumulates a considerable body of literature [6], [7], [8], [9].

As mentioned above, the main idea of AD methods consists in building a model of the norm and using this models to detect deviations. Traditional methods for AD include One-class Support Vector Machines (OSVM) [10], Support Vector Data Description (SVDD) [11], Kernel Density Estimation (KDE) [12], Isolation Forests (IF) [13]. These approaches have been applied to detect network traffic anomaly detection [3], [14], [15]. For example, Tran et al. [14] extracted statistics from network traffic, such as the number of TCP packets, the average packet size, and applied OSVM on these extracted statistics to detect anomalous network packets. More recently, Tao et al. [15] proposed a network traffic anomaly detector based on IF. Despite their effectiveness to handle some datasets with hand-crafted features, the performance of these approaches drastically degrades with complex high-dimensional data [16]. Besides, they suffer from poor computational scaling with input dataset size, due to the quadratic definition of kernels.

To cope with these limitations, hybrid approaches combine a dimensionality reduction approach (e.g., autoencoder) with a classical anomaly detector (e.g., OSVM) [17]. While the former reduces the dimension of the input data, the latter uses this compressed representation to discriminate between regularities and anomalies. [18] proposed, Auto-IF, an approach for network intrusion detection that combines AEs and IF. However, in these approaches, the dimensionality reduction phase is independent from the task of anomaly detection, which does not guarantee the relevance of the final representation [19].

In recent years, many approaches directly adapt dimensionality reduction algorithms for anomaly detection, by considering the reconstruction error as an anomaly score [20]. Dimensionality-based anomaly detection assumes that nominal data can be projected in an optimal low-dimensional subspace, while anomalous instances cannot. Consequently, the reconstructions of normal instances are close to the original data, with a low reconstruction error, while that of outliers are discrepant. In particular, autoencoders are commonly used in network anomaly detection, such as in the proposition from Chen et al. [21].

However, the performance of these approaches depends on the availability of anomaly-free training data. The presence of some anomalies in the training instances creates a bias in the learned representation of the normality, and increases the False Positive Rate (FPR) [22]. In this case, AEs learn to reconstruct both nominal and anomalous data, and fail to discriminate them. In a real-world environment, there is no guarantee that training data are entirely clean, and some unusual abnormal events may be hidden in the collected data, for instance due to malfunctioning sensors, adversarial attacks, or simply cluttered erroneous measurements. This concern has been repeatedly reported in the anomaly detection community [22], [23], which advocates the use of robust unsupervised anomaly detectors that are insensitive to anomalies contaminating ground-truth data.

Inspired by the Robust PCA (RPCA) [24], Robust Deep Autoencoders (RDA) [25], [26] extend AEs to the case where the training data may be corrupted by some unlabeled anomalies. RDAs assume that the training anomalies cannot be projected into the exact same subspace of the nominal data. Consequently, RDAs filter training instances that are difficult to reconstruct, and learn an accurate representation of the remaining training data. More precisely, RDAs split the input data matrix X into two parts: $X = L_D + S$, where L_D contains nominal data, and S contains anomalies. RDA minimizes the following objective:

$$\min_{E,D} \|L_D - D(E(L_D))\|_2 + \lambda \|S\|_0 \quad (1)$$
$$subject\ to \quad L_D + S = X$$

where E denotes the encoding function from the input to the hidden space, D is the decoding map from the hidden layer to the output, and λ is a hyperparameter that controls the trade-off between the reconstruction error and the sparsity of S. The sparsity is induced by regularizing S with the l_0 norm, which represents the number of non-zero elements in this matrix. Optimizing this objective function (1) is not tractable, due to the presence of the l_0 norm. To cope with this problem, and to filter training outliers, Zhou et al. [25] proposed to replace the l_0 norm with the $l_{2,1}$ norm, which is defined as:

$$\|X\|_{2,1} = \sum_j \|x_j\| = \sum_j \sqrt{\sum_i |x_{ij}|^2} \quad (2)$$

The $l_{2,1}$ norm consists in applying the l_2 norm over each column of the data matrix, and group all these results using the l_1 norm. Outliers are filtered by applying the $l_{2,1}$ norm on S^T

$$\min_{E,D} \|L_D - D(E(L_D))\|_2 + \lambda \left\|S^T\right\|_{2,1} \quad (3)$$
$$subject\ to \quad L_D + S = X$$

Recently, Kotani et al. [27] proposed an RDA-based approach for network flow intrusion detection, which proved

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

to reduce the number of false positives on real-world traffic datasets. However, the performance of RDAs is highly dependent on the hyperparameter λ. A low λ results in rejecting too many training data in S, while a high λ over-penalizes data isolated in S, and consequently, enforces the RDA to reconstruct all training data. For this reason, the value of λ needs to be chosen carefully with a dedicated exploratory strategy that implicitly requires labelled anomalies. Thus, such RDAs cannot deliver reliable results in uncontrolled environments that require unsupervised approaches. This is why we propose RADON, which addresses this important limitation, by redefining the reconstruction problem.

III. Method

In the following, we detail three contributions. First, we introduce the new objective to optimize, based on the cosine function. Then, we revise the traditional projection constraint: instead of decomposing the input matrix X into one low-matrix L and a sparse matrix S, we propose to divide the set $\mathbb{X}$ into three distinct subsets, $\mathbb{L}_\mathbb{D}$, $\mathbb{S}$, and $\mathbb{U}$, respectively composed of the estimated normal, abnormal, and undetermined samples. Finally, we explain how to constitute these subsets thanks to a unimodal triangular thresholding [28] on the reconstruction scores histogram.

Inspired by some metric learning approaches [29], [30], our objective is to learn a robust projection, where nominal data are similar to their reconstruction, but also where anomalies are explicitly badly reconstructed. Indeed, RDAs learn a non-linear representation of the nominal class by filtering training anomalies, but do not leverage them to increase the relevance of this representation for the task of anomaly detection. RDA reconstruction scores are based on the l_2 norm, which is not bounded. That is why we propose an objective based on the cosine similarity function, with values in the range [0,1]. More precisely, we propose to train the RDA such that training inliers are collinear to their reconstructions, and training anomalies and their reconstructions are orthogonal.

We also propose to adapt the projection strategy of the $l_{2,1}$ RDA: instead of decomposing the input matrix into two matrices L and S, we divide the input data into three disjoint subsets $\mathbb{X} = \mathbb{L}_\mathbb{D} \cup \mathbb{S} \cup \mathbb{U}$. The first subset $\mathbb{L}_\mathbb{D}$ represents data points that are colinear to their reconstructions, with $cos(\mathbb{L}_\mathbb{D}, D(E(\mathbb{L}_\mathbb{D})))$ close to 1. The second subset $\mathbb{S}$ represents data that cannot be reconstructed, with $cos(\mathbb{S}, D(E(\mathbb{S})))$ close to 0. The third subset $\mathbb{U}$ is formed of the remaining points, with a cosine score around 0.5. For these instances, it is not possible to confidently determine whether they should be correctly reconstructed or not. Indeed, as we are jointly learning to reconstruct inliers and to seclude outliers, we need to consider only confident accepted or rejected instances, with a cosine score very close to 1 or to 0, respectively. Considering in-between instances may progressively destabilize the training.

We reformulate the optimization problem as follows:

$$\min_{E,D} \; (1 - cos(\mathbb{L}_\mathbb{D}, D(E(\mathbb{L}_\mathbb{D}))))^2 + (0 - cos(\mathbb{S}, D(E(\mathbb{S}))))^2 \quad (4)$$

$$subject\ to \;\; \mathbb{X} = \mathbb{L}_\mathbb{D} \cup \mathbb{S} \cup \mathbb{U}$$

The training strategy is iterative and alternates between updating the AE parameters and updating the subsets $\mathbb{X} = \mathbb{L}_\mathbb{D} \cup \mathbb{S} \cup \mathbb{U}$. We optimize our objective function for a fixed number of epochs, we project the data according to the constraint, and we repeat this process using the updated values (see Figure 1).

The convergence criterion consists in that all training instances are classified whether in $\mathbb{S}$ or $\mathbb{L}_\mathbb{D}$, and no instances remain unclassified in $\mathbb{U}$. We stop the training when $\frac{\|\mathbb{U}\|_2}{\|\mathbb{X}\|_2} < \epsilon$. As $\{\mathbb{S} \cup \mathbb{L}_\mathbb{D}\}$ may converge to a fixed set that is different from $\mathbb{X}$, we define a maximum number of epochs, after which we stop the training even if this inequality is not verified.

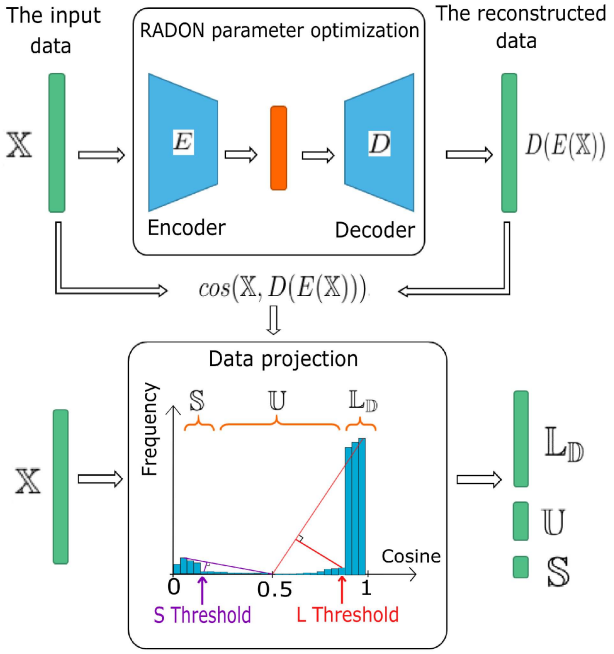

Fig. 1: RADON training strategy, which alternates between updating the AE parameters (top) and updating the subsets $\mathbb{X} = \mathbb{L}_\mathbb{D} \cup \mathbb{S} \cup \mathbb{U}$ (bottom).

To decompose $\mathbb{X}$ into the three subsets, we propose to form a histogram with reconstruction values $cos(\mathbb{X}, D(E(\mathbb{X})))$, enabling the use of image segmentation thresholding techniques [31]. We assume an unbalanced dataset, where normal data are over-represented compared to anomalies. The resulting reconstruction histogram should present a dominant peak, and may or may not have a second, much smaller peak. Consequently, we propose to use the unimodal thresholding [28], which specifically addresses this family of histograms. The unimodal threshold is performed by drawing a line between the peak of the histogram, P, with coordinates [centre of the main bin; peak frequency]; and the first minimum, Q, with coordinates [centre of the first empty bin, 0] (see Figure 2 (left)). We select the bin centre whose peak is farthest from this line as the threshold.

Initially, $\mathbb{L}_\mathbb{D} = \mathbb{X}$, and both $\mathbb{U}$ and $\mathbb{S}$ are empty. For the first iteration, we minimize the objective function (4). Then,

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

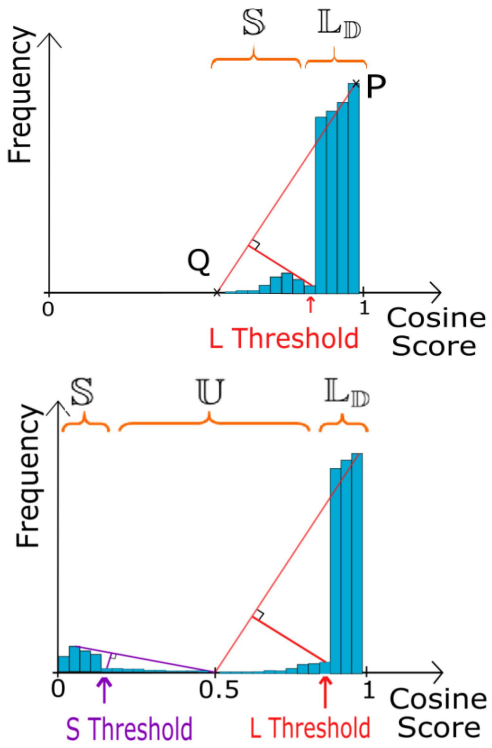

Fig. 2: The thresholding strategy at the end of the first iteration (top), and at the end of the following ones (bottom).

we compute the reconstruction error of the input data, and we split it into two parts, according to the triangular threshold (see Figure 2 (top)): instances having a reconstruction score greater than the triangular threshold are put in $\mathbb{L}_\mathbb{D}$, and the remaining data are placed in $\mathbb{S}$. Then, we continue RADON training using both positive and negative data. At the end of the next iteration, we have a reconstruction histogram similar to Figure 2 (bottom). We divide the histogram into two histograms: the first histogram, H_L, containing instances with a cosine score greater than 0.5, and the second histogram, H_S, containing instances with cosine less than 0.5. We threshold both histograms using the unimodal threshold (see Figure 2 (bottom)). Instances with a cosine greater than L threshold are classified in $\mathbb{L}_\mathbb{D}$, instances with a cosine less than S threshold are in $\mathbb{S}$, and the remaining are in $\mathbb{U}$.

Finally, as we threshold the reconstruction histogram to reject outliers, the number of bins is an important hyperparameter that plays a key role in this approach. While a small number of bins may reduce the impact of noise caused by the randomness of sampling, a large number of bins provide more details about the original data. This hyperparameter can be fine-tuned, where the optimal value with the best performance on the validation sets is selected. However, this approach is computationally expensive, as it involves repeating the training several times for different bin number values. Many heuristics were developed to estimate the optimal number of bins from the underlying data [32].

Let n be the number of data points in the original data, h the bin width, and k the optimal number of bins. The Rice rule estimates $k = 2\sqrt[3]{n}$. This estimator considers only the volume of the data n. Other rules not only depend on n but also on the data variability. Scott's rule defines $h = 3.491\sigma n^{-\frac{1}{3}}$, where σ is the standard deviation of samples. As σ is not robust to outliers, Freedman and Diaconis proposed a robust estimator, defined as $h = 2IQRn^{-\frac{1}{3}}$, where IQR refers to the interquartile range. However, these methods generally make strong assumptions about the underlying data distribution, and can be sub-optimal if these assumptions are not verified. Recently, Knuth [33] proposed a non-parametric data-based approach to determine the optimal number of bins, with no assumption on data distribution. This approach relies on the Bayesian framework to infer the posterior probability of the number of bins in a piecewise-constant density function. In our experiments, we compare the performance of RADON with 3 different bin estimators: (1) the Rice rule, (2) the Freedman and Diaconis (FD) estimator, and (3) the Knuth's estimator.

IV. Experiments

In this section, we compare our approach, RADON, against several unsupervised anomaly detectors. Particularly, we consider the Isolation Forest (IF) [13], OCSVM with a Gaussian kernel, which is identical to SVDD, vanilla AEs, $l_{2,1}$ RDA [25], and RVAE [34].

We use three real-world datasets from two different fields: computer vision and network traffic anomaly detection. Firstly, we evaluate the performance of our approach using the well-known computer vision dataset: the MNIST dataset. In particular, we investigate the robust rejection strategy of RADON, by displaying both inliers and rejected outliers at the end of the training. Secondly, we experiment on one of the most popular datasets in network intrusion detection: the NSL-KDD. Finally, further experiments are run on a recent IoT dataset: the MedBIoT dataset. In the following, we introduce these datasets, the training protocol, and the experimental results.

A. Dataset Description

1) MNIST Dataset: To assess the performance of the $l_{2,1}$ RDA in anomaly detection, Zhou and Paffenroth used a subset of the MNIST dataset [25]. MNIST is a popular computer vision dataset of handwritten digits. Each instance in this dataset is a 28x28-pixel image that represents one digit between 0 and 9. In their experiments, they consider images of digit 4 as normal data, while images of other digits are considered outliers. From the original MNIST dataset, they randomly selected a subset of 4859 images of the digit 4, and 265 images of other digits (e.g. 0, 2, 3). Here, outliers represent 5.2% of this dataset.

2) NSL-KDD Dataset: The second dataset used in our experiments is the NSL-KDD dataset [36]. The NSL-KDD is a benchmark dataset widely used to assess the performance of IDSs in distinguishing malicious network connections, also known as "intrusions", and normal ones. This dataset is divided into two subsets: the training subset with 125973

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I: Extracted flow features using NFStream. See [35] for detailed feature descriptions.

Features			Abbreviations
src_port	src2dst_stdev_piat_ms	src2dst_duration_ms	src : source (e.g., src_port means the source port of the packet)
dst_port	src2dst_max_piat_ms	src2dst_packets	dst : destination
protocol	dst2src_min_piat_ms	src2dst_bytes	src2dst : traffic from source to destination
ip_version	dst2src_mean_piat_ms	bidirectional_min_piat_ms	piat : packet inter arrival time.
dst2src_stdev_piat_ms	dst2src_max_piat_ms	bidirectional_mean_piat_ms	stdev : standard deviation
bidirectional_duration_ms	bidirectional_syn_packets	bidirectional_stdev_piat_ms	ps : packet size
src2dst_mean_piat_ms	bidirectional_max_piat_ms	bidirectional_packets	syn, urg, ack, psh, fin rst, ece, and cwr are TCP flags
bidirectional_cwr_packets	bidirectional_bytes	bidirectional_ece_packets	
bidirectional_urg_packets	bidirectional_ack_packets	src2dst_syn_packets	
bidirectional_psh_packets	bidirectional_rst_packets	bidirectional_fin_packets	
dst2src_mean_ps	dst2src_stdev_ps	dst2src_max_ps	
src2dst_cwr_packets	dst2src_duration_ms	src2dst_ece_packets	
bidirectional_max_ps	src2dst_min_ps	src2dst_mean_ps	
dst2src_cwr_packets	dst2src_ece_packets	dst2src_urg_packets	
dst2src_syn_packets	src2dst_max_ps	dst2src_ack_packets	
dst2src_min_ps	dst2src_psh_packets	src2dst_stdev_ps	
dst2src_rst_packets	dst2src_fin_packets	src2dst_min_piat_ms	
dst2src_packets	src2dst_urg_packets	dst2src_bytes	
src2dst_ack_packets	bidirectional_min_ps	src2dst_psh_packets	
bidirectional_mean_ps	src2dst_rst_packets	bidirectional_stdev_ps	
src2dst_fin_packets			

data points and the test subset with 22544 records. Normal records represent 53% of the training data. Each instance of this dataset contains 41 features extracted from the network traffic, e.g., protocol type, TCP flags, and labeled as normal or as a threat. This dataset encompasses 39 types of attacks, with 17 only present in the test subset. These attacks belong to 4 categories : Denial of Service (DoS), Remote to Local (R2L), User to Root (U2R), and Probes. We rescale numeric features to be in the range $[0, 1]$. Categorical features are one-hot encoded.

3) MedBIoT Dataset: The MedBIoT dataset [37] is an IoT dataset collected from a medium-sized IoT network that contains 83 IoT devices, belonging to four categories: switches, light bulbs, locks, and fans. Three botnet malware were deployed, Mirai, Bashlite, Torii attacks. Both normal and malicious raw network traffic were collected for each device-type and made public. Globally, this dataset comprises over 17 million network packets: around 30% of this traffic is anomalous and the remaining 70% is benign. Two reasons justify our choice of this dataset: (i) firstly, unlike other IoT datasets that focus on small-sized networks, MedBIoT provides raw traffic data collected from a large network containing 83 IoT devices, (ii) secondly, this dataset is recently published, in 2020, and contains up-to-date IoT botnet traffic. We extract network header metadata from these raw packets, using the open-source framework NFStream [35]. In particular, we extract 61 flow-based features, which are detailed in Table I.

We randomly split the benign data into 60% for the training, 20% for the validation, and 20% for testing. Then, we contaminate the benign training data, so that anomalies represent 10% of the final number of training instances. These contaminants are randomly selected from all anomalous classes of MedBIoT. The remaining anomalies are randomly split into 50% validation and 50% test. In this dataset, the cardinality of categorical features is high. To handle the problem of high dimensional categorical data encoding, and similar to [38], we use CountEncoder [39] which encodes categorical features with their frequency of occurrence. Indeed, categorical features are normalized using Min-Max normalization method and encoded using Count Encoder.

B. Training Protocol And Parameter Settings

We use 10-fold cross-validation, to provide a reliable estimation of model performance, and to limit the impact of model parameter random initialization. In all experiments, we use symmetric autoencoders, with the standard Multi-layer Perceptron (MLP) feed-forward architectures For the MNIST experiments, we use the same architecture proposed by Zhou and Paffenroth [25]. The encoder is constituted of two feedforward layers, having 400 and 200 artificial neurons, respectively. In the NSL-KDD experiments, the encoder is composed of a eight-neuron single feedforward layer. In the MedBIoT experiments, the encoder is a 3-layer MLP with 61-16-4 units. For the competing methods, we fine-tune their hyperparameters on the validation subset. We select the optimal hyperparameters that maximize validation Area Under the Curve of the Receiver Operating Characteristics (AUROC) and Area Under the Precision-Recall Curve (AUPRC). For the MNIST experiments, each iteration contains 50 epochs, while this projection occurs every 100 epochs for NSL-KDD and MedBIoT experiments. We use a learning rate of 0.001, an empirical maximum number of epochs of 800, and a batch size of 256. Finally, all the experiments were run on a laptop equipped with a 12-core Intel i7-9850H CPU clocked at 2.6GHz, with 16GB of RAM memory and with NVIDIA Quadro P2000 GPU.

C. Experimental Results And Discussion

In our experiments, we use two commonly used evaluation metrics in AD: the AUROC, and the AUPRC.

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

TABLE II: MNIST experimental results. * indicates that the result is significant, according to Welch's test with a p-value=0.05

Metric	Methods							
	IF	OSVM	Vanilla AE	RVAE	$l_{2,1}$ RDA	RADON		
						FD	Rice	Knuth
AUROC	77.8 ± 0.9	82.5 ± 0.7	89.4 ± 3.8	90.4 ± 0.9	91.0 ± 0.5	96.3 ± 0.7	$\mathbf{97.0 \pm 0.4}$*	$\mathbf{97.0 \pm 0.5}$*
AUPRC	46.0 ± 0.9	51.6 ± 0.7	49.7 ± 2.0	63.7 ± 1.0	63.2 ± 1.5	70.4 ± 8.3	74.6 ± 4.4	$\mathbf{75.5 \pm 4.2}$*

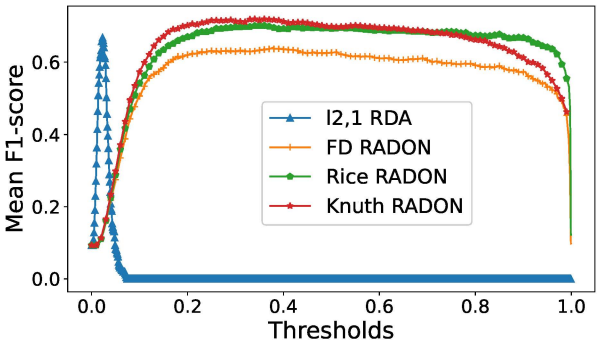

Fig. 3: Variation of the mean F1-scores of the $l_{2,1}$ RDA and RADON with different bin estimators, on the MNIST.

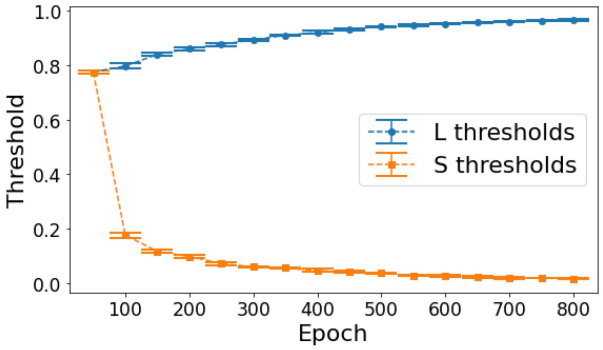

Fig. 4: Variation of $\mathbb{L}_{\mathbb{D}}$ and $\mathbb{S}$ thresholds during the training stage

1) MNIST Experimental Results: The results are reported in Table II. Our approach RADON outperforms all baselines on the MNIST dataset. We start by comparing RADON with the $l_{2,1}$ RDA. Globally, our proposed approach, trained with the unimodal threshold and with different histogram bin estimators, performs notably better than $l_{2,1}$ RDA. RADON with Knuth's estimator improves the mean value of $l_{2,1}$ RDA AUROC and AUPRC, by 6% and 12%, respectively. We note that RADON reports a larger AUPRC confidence interval than $l_{2,1}$ RDA. We use Welch's test to assess the significance of these results, and all these results are statistically significant, with p-value = 0.05. Moreover, RADON outperforms other anomaly detectors, IF, OSVM, and vanilla AE.

Besides, we report a slight advantage of RADON with Knuth's bin estimator over other estimators, mainly regarding the AUPRC metric. This can be explained by the fact that the number of bins estimated with Knuth's estimator is adapted to the original data, with no prior on the underlying distribution. Consequently, the corresponding histogram is more representative.

We compare in Figure 3 the F1-score variation of the $l_{2,1}$ RDA and RADON, for different thresholds. The mean F1-score of $l_{2,1}$ RDA is characterized by a sharp peak, reached at 0.023, and the performs degrades abruptly when the threshold is above or below 0.023. However, the mean F1-score of RADON, with different bin estimators, is marked by a plateau for a large range of thresholds. This observation can be attributed to our reformulation of the training objective, which jointly maximizes the distance between the reconstruction errors of inliers and outliers, and makes the classification less sensitive to the threshold.

To validate this interpretation, we inspect the variation of both $\mathbb{L}_{\mathbb{D}}$ and $\mathbb{S}$ thresholds after each iteration during the training (see Figure 4). After the first projection at 50 epochs, $\mathbb{L}_{\mathbb{D}}$ and $\mathbb{S}$ thresholds are equal. The filtering becomes more and more parsimonious and selective, as $\mathbb{L}_{\mathbb{D}}$ thresholds are moving iteratively closer to 1, and $\mathbb{S}$ thresholds get closer to 0. Consequently, this variation explains the plateau reported in Figure 3, as the discrepancy between both inliers and outliers reconstruction error is increasing after each projection, and maximized at the end of the training.

Figure 5 illustrates an example of the training data used in one fold of 10-fold cross-validation. On the left, we depict training outliers rejected by RADON. The majority of $\mathbb{S}$ is composed of outliers belonging to different classes of digits, alongside a few false positives. On the right, we find nominal instances. Almost all these images are 4 digits. It is worth mentioning that, empirically, training outliers not rejected by RADON belong to classes 1, 7, and 9. These digits can be found within the digit 4. Thus, the latent space of those digits is close to the one for the Digit 4 class, easing the reconstruction task of the autoencoder, and introducing false positives.

This observation can be explained by the natural similarity between these digits and digit 4.

2) NSL-KDD Experimental Results: In this section, we present the experimental results reported on the NSL-KDD. To study the sensitivity of anomaly detectors with respect to the ratio of anomaly contamination, we vary the training anomaly percentage. We prepare three training subsets, containing 1%, 5%, and 10% of outliers, respectively. Anomaly instances are selected randomly from all classes of NSL-KDD training anomalies. We corrupt training nominal data by mixing it with these outliers and we randomly shuffle all training data. To

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

TABLE III: NSL-KDD experimental results, with different percentages γ of training outliers. * indicates that the result is significant, according to Welch's test with a p-value=0.05.

γ	Metric	Methods							
		IF	OSVM	Vanilla AE	RVAE	$l_{2,1}$ RDA	RADON		
							FD	Rice	Knuth
1%	AUROC	83.6 ± 0.5	81.6 ± 0.1	91.6 ± 1.0	90.8 ± 0.1	93.7 ± 0.7	94.2 ± 0.9	$\mathbf{94.8 \pm 0.8}$*	94.7 ± 0.8
	AUPRC	90.8 ± 0.5	89.6 ± 0.1	91.5 ± 1.5	90.2 ± 0.1	92.2 ± 0.9	93.7 ± 2.0	94.1 ± 1.4	94.4 ± 1.3
5%	AUROC	84.1 ± 0.4	84.4 ± 0.1	89.5 ± 0.8	90.3 ± 1.3	91.9 ± 0.7	94.5 ± 0.5	94.4 ± 0.6	$\mathbf{95.2 \pm 1.2}$*
	AUPRC	90.9 ± 0.3	90.3 ± 0.1	89.7 ± 1.2	89.4 ± 1.4	91.0 ± 0.7	94.2 ± 1.2	94.4 ± 1.0	$\mathbf{95.6 \pm 1.1}$*
10%	AUROC	82.7 ± 0.5	83.2 ± 0.1	88.6 ± 0.7	90.1 ± 0.4	91.0 ± 0.6	93.8 ± 1.4	$\mathbf{94.4 \pm 1.1}$*	$\mathbf{94.4 \pm 1.1}$*
	AUPRC	90.5 ± 0.5	89.7 ± 0.1	88.9 ± 1.0	90.5 ± 0.7	91.0 ± 0.9	94.0 ± 1.5	94.2 ± 1.6	$\mathbf{94.6 \pm 1.2}$*

TABLE IV: MedBIoT experimental results, with $\gamma = 0.1$. * indicates that the result is significant, according to Welch's test with a p-value=0.05.

Device	Metric	Methods						
		IF	OSVM	Vanilla AE	$l_{2,1}$ RDA	RADON		
						FD	Rice	Knuth
Lock	AUROC	93.0 ± 1.5	92.0 ± 0.1	74.2 ± 2.0	95.6 ± 0.5	99.2 ± 2.8	$\mathbf{99.3 \pm 0.3}$*	$\mathbf{99.3 \pm 2.0e{-2}}$*
	AUPRC	95.9 ± 0.7	95.3 ± 0.4	80.6 ± 1.1	94.9 ± 0.7	99.5 ± 1.7	99.5 ± 0.2	$\mathbf{99.6 \pm 1.7e{-2}}$*
Fan	AUROC	$94.1 \pm 4.2e{-2}$	$94.0 \pm 9.2e{-2}$	98.9 ± 0.4	$99.5 \pm 2.5e{-2}$	$\mathbf{99.9 \pm 1.3e{-2}}$*	$\mathbf{99.9 \pm 1.3e{-2}}$*	$\mathbf{99.9 \pm 2.8e{-2}}$*
	AUPRC	$97.9 \pm 3.2e{-3}$	$97.8 \pm 3.2e{-2}$	93.7 ± 1.8	$99.3 \pm 5.4e{-2}$	$\mathbf{99.9 \pm 2.9e{-2}}$*	$\mathbf{99.9 \pm 2.8e{-2}}$*	$\mathbf{99.9 \pm 3.4e{-2}}$*
Bulbs	AUROC	$94.0 \pm 5.3e{-2}$	$94.0 \pm 6.3e{-2}$	84.5 ± 0.6	$99.5 \pm 2.0e{-2}$	$\mathbf{99.9 \pm 2.1e{-2}}$*	$\mathbf{99.9 \pm 2.1e{-2}}$*	$\mathbf{99.9 \pm 2.8e{-2}}$*
	AUPRC	$97.7 \pm 2.3e{-3}$	$97.7 \pm 2.3e{-2}$	89.7 ± 3.2	$99.2 \pm 4.7e{-2}$	$\mathbf{99.9 \pm 1.3e{-2}}$*	$\mathbf{99.9 \pm 1.4e{-2}}$*	$\mathbf{99.9 \pm 4.7e{-2}}$*
Switch	AUROC	$94.1 \pm 7.6e{-2}$	$94.1 \pm 7.1e{-2}$	92.5 ± 0.4	$99.4 \pm 1.2e{-2}$	$\mathbf{99.8 \pm 0.9e{-2}}$*	$\mathbf{99.8 \pm 0.1e{-2}}$*	$\mathbf{99.8 \pm 0.7e{-2}}$*
	AUPRC	$97.2 \pm 4.1e{-3}$	$97.2 \pm 3.2e{-2}$	93.4 ± 2.4	$99.0 \pm 4.4e{-2}$	$\mathbf{99.8 \pm 4.8e{-2}}$*	$\mathbf{99.8 \pm 4.8e{-2}}$*	$\mathbf{99.8 \pm 4.7e{-2}}$*

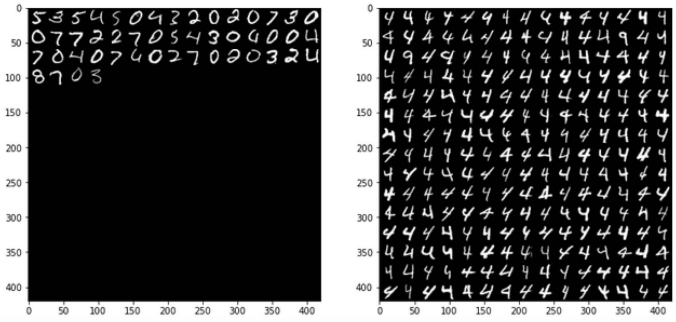

Fig. 5: (left) Training images rejected by our approach, RADON, and considered as outliers $\mathbb{S}$. (right) Training images considered as nominal instances $\mathbb{L}_{\mathbb{D}}$.

have a fair comparison and to use the same data for different model training, we fix a seed for data shuffle and training anomaly instance selection.

Table III summarizes these results. RADON shows globally superior results, for different percentages γ of training outliers. This demonstrates that RADON is very competitive, compared to robust anomaly detection methods. Particularly, we focus on the comparison between RADON and $l_{2,1}$ RDA. Two key findings emerge. Firstly, in the case of a low ratio of training outliers, i.e., 1% of the training data, RADON with the different bin estimators outperforms $l_{2,1}$ RDA. We report a 1% and 2.2% points increase for mean AUROC and AUPRC, respectively, with Knuth's estimator. Secondly, when the ratio of training outliers becomes relatively important, representing 5% or 10% of the training data, the difference between both approaches is more noticeable, with around 3% and 4% points better mean AUROC and AUPRC, respectively. The best results are reported with Knuth's estimator.

3) MedBIoT Experimental Results: The third experiment aims to validate the proposed approach in IoT traffic anomaly detection, using the MedBIoT dataset. The results are reported in Table IV. Our contribution, RADON, shows the best performance for all device-types, compared to other baselines. RADON reports an AUROC and an AUPRC higher than 0.99 for all device-types, and with the different estimators. In particular, we note the robustness of our contribution compared to vanilla AE. While vanilla AE performance is considerably impacted by training contaminants, RADON does not learn to reconstruct outliers and yields good results. Indeed, RADON exceeds vanilla AUROC by 14% and vanilla AUPRC by 10%, on average. Furthermore, RADON performs slightly better than $l_{2,1}$ RDA.

V. Conclusion

In this paper, we introduced RADON, a robust autoencoder for unsupervised outlier detection. Our approach filters training outliers, and leverages them to learn a robust projection, where inliers and their reconstructions are similar, while outliers are poorly reconstructed. To effectively filter training outliers, we performed unimodal thresholding of the reconstruction error histogram. Furthermore, we reformulated the training strategy as a metric learning problem. Experimental evaluation of RADON on the MNIST, the NSL-KDD, and the MedBIoT benchmark datasets showed competitive results, both in image and network traffic anomaly detection. In the future, we will investigate the histogram thresholding stage. We aim to make

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

this phase more robust to the histogram smoothness, which is parameterized here by the number of bins. Besides, we will extend this approach to take into account the temporal dependencies.

REFERENCES

[1] D. Evans, "How the Next Evolution of the Internet Is Changing Everything," p. 11, 2011.

[2] "As malware and network attacks increase in 2019, zero day malware accounts for 50% of detections," Dec. 2019. [Online]. Available: https://www.helpnetsecurity.com/2019/12/13/network-attacks-2019/

[3] G. Fernandes, J. J. P. C. Rodrigues, L. F. Carvalho, J. F. Al-Muhtadi, and M. L. Proença, "A comprehensive survey on network anomaly detection," *Telecommunication Systems*, vol. 70, no. 3, pp. 447–489, Mar. 2019.

[4] D. Hawkins, *Identification of Outliers*, ser. Monographs on applied probability and statistics, 1980.

[5] Y. Le Cun and F. Fogelman-Soulié, "Modèles connexionnistes de l'apprentissage," *Intellectica*, vol. 2, no. 1, pp. 114–143, 1987.

[6] X. Xie, C. Wang, S. Chen, G. Shi, and Z. Zhao, "Real-time illegal parking detection system based on deep learning," in *Proceedings of the 2017 International Conference on Deep Learning Technologies*. Association for Computing Machinery, 2017, pp. 23–27.

[7] A. O. Adewumi and A. A. Akinyelu, "A survey of machine-learning and nature-inspired based credit card fraud detection techniques," *International Journal of System Assurance Engineering and Management*, vol. 8, no. 2, pp. 937–953, 2017.

[8] D. Kwon, H. Kim, J. Kim, S. C. Suh, I. Kim, and K. J. Kim, "A survey of deep learning-based network anomaly detection," *Cluster Computing*, vol. 22, no. 1, pp. 949–961, 2019.

[9] S. Thudumu, P. Branch, J. Jin, and J. J. Singh, "A comprehensive survey of anomaly detection techniques for high dimensional big data," *Journal of Big Data*, vol. 7, no. 1, p. 42, 2020.

[10] B. Schölkopf, R. Williamson, A. Smola, J. Shawe-Taylor, and J. Platt, "Support vector method for novelty detection," in *NIPS*. MIT Press, 1999, pp. 582–588.

[11] D. M. Tax and R. P. Duin, "Support vector data description," *Machine Learning*, vol. 54, no. 1, pp. 45–66, 2004.

[12] E. Parzen, "On estimation of a probability density function and mode," 1962.

[13] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE ICDM*. IEEE, 2008, pp. 413–422.

[14] Q.-A. Tran, H. Duan, and X. Li, "One-class Support Vector Machine for Anomaly Network Traffic Detection," *China Education and Research Network (CERNET), Ts-inghua University*, p. 7, 2004.

[15] X. Tao, Y. Peng, F. Zhao, P. Zhao, and Y. Wang, "A parallel algorithm for network traffic anomaly detection based on Isolation Forest," *International Journal of Distributed Sensor Networks*, p. 11, 2018.

[16] L. Ruff, R. A. Vandermeulen, N. Görnitz, A. Binder, E. Müller, K.-R. Müller, and M. Kloft, "Deep semi-supervised anomaly detection," *ICLR*, p. 23, 2020.

[17] P. Perera and V. M. Patel, "Learning deep features for one-class classification," *IEEE Transactions on Image Processing*, vol. 28, no. 11, pp. 5450–5463, 2019.

[18] K. Sadaf and J. Sultana, "Intrusion Detection Based on Autoencoder and Isolation Forest in Fog Computing," *IEEE Access*, vol. 8, pp. 167 059–167 068, 2020.

[19] L. Ruff, R. A. Vandermeulen, N. Görnitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, and M. Kloft, "Deep one-class classification," in *ICML*, vol. 80, 2018.

[20] J. Chen, S. Sathe, C. Aggarwal, and D. Turaga, "Outlier detection with autoencoder ensembles," in *SDM*, 2017, pp. 90–98.

[21] Z. Chen, C. K. Yeo, B. S. Lee, and C. T. Lau, "Autoencoder-based network anomaly detection," Apr. 2018, pp. 1–5.

[22] H. Ringberg, A. Soule, J. Rexford, and C. Diot, "Sensitivity of PCA for traffic anomaly detection," *ACM SIGMETRICS Performance Evaluation Review*, vol. 35, no. 1, pp. 109–120, Jun. 2007.

[23] C. O'Reilly, A. Gluhak, and M. A. Imran, "Distributed Anomaly Detection Using Minimum Volume Elliptical Principal Component Analysis," *IEEE Transactions on Knowledge and Data Engineering*, vol. 28, no. 9, pp. 2320–2333, Sep. 2016, conference Name: IEEE Transactions on Knowledge and Data Engineering.

[24] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *Journal of the ACM*, vol. 58, no. 3, pp. 1–37, 2011.

[25] Zhou and R. C. Paffenroth, "Anomaly detection with robust deep autoencoders," in *Proceedings of the 23rd ACM SIGKDD - KDD '17*, 2017, pp. 665–674.

[26] R. Chalapathy, A. K. Menon, and S. Chawla, "Robust, deep and inductive anomaly detection," in *Machine Learning and Knowledge Discovery in Databases*, 2017.

[27] G. Kotani and Y. Sekiya, "Unsupervised Scanning Behavior Detection Based on Distribution of Network Traffic Features Using Robust Autoencoders," in *2018 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2018.

[28] P. L. Rosin, "Unimodal thresholding," *Pattern Recognition*, vol. 34, no. 11, pp. 2083–2096, 2001.

[29] G. Lefebvre and C. Garcia, "Learning a bag of features based nonlinear metric for facial similarity," in *IEEE International Conference on Advanced Video and Signal Based Surveillance*, 2013, pp. 238–243.

[30] S. Berlemont, G. Lefebvre, S. Duffner, and C. Garcia, "Class-balanced siamese neural networks," *Neurocomputing*, vol. 273, pp. 47–56, 2018.

[31] M. Waseem Khan, "A survey: Image segmentation techniques," *International Journal of Future Computer and Communication*, pp. 89–93, 2014.

[32] P. Sulewski, "Equal-bin-width histogram versus equal-bin-count histogram," 2020.

[33] K. H. Knuth, "Optimal data-based binning for histograms and histogram-based probability density models," *Digital Signal Processing*, vol. 95, 2019.

[34] H. Akrami, S. Aydore, R. Leahy, and A. A. Joshi, "Robust variational autoencoder for tabular data with β divergence," *ICML UDL*, 2020.

[35] "NFStream - a Flexible Network Data Analysis Framework." [Online]. Available: https://nfstream.org/

[36] (2009) NSL-KDD dataset UNB. [Online]. Available: https://www.unb.ca/cic/datasets/nsl.html

[37] A. Guerra-Manzanares, J. Medina-Galindo, H. Bahsi, and S. Nõmm, "MedBIoT: Generation of an IoT Botnet Dataset in a Medium-sized IoT Network:," in *Proceedings of the 6th International Conference on Information Systems Security and Privacy*, Valletta, Malta, 2020, pp. 207–218.

[38] C. Mougan, D. Masip, J. Nin, and O. Pujol, "Quantile encoder: Tackling high cardinality categorical features in regression problems," in *Modeling Decisions for Artificial Intelligence*, 2021, pp. 168–180.

[39] "Count encoder." [Online]. Available: https://contrib.scikit-learn.org/category$_e ncoders/count.html$

Authorized licensed use limited to: Universita della Calabria. Downloaded on January 19,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.